Example:

```
#include <iostream>
using namespace std;
class Student {
    public:
        string name;
        int age;
        void show () {
            cout << name << " " << age;
        }
};
```

Student → class
name, age → data members (variables)
show () → members function.

```
int main () {
    Student s1;
    s1.name = "Krish";
    s1.age = 19;
    s1.show();
}
```

Krish 19
s1 → object of class Student
we access members using . (dot operator)

What are access specifiers?
Access specifiers control who can access the class members.
There are three types:

1. public
2. private.
3. protected.

int sum(int a, int b)
float sum(int a, int b); // Error. } Because the compiler do not decide which function to call.

Practice problems: Smart Number Analyzer.

- If one parameter given find even, odd.
- If two parameter given find max, min, which is greater.
- Prime number in range.

OOPS

#Day - 12 (Object oriented Programming)

OOP in CPP is a programming style where we organize code using object and classes instead of only function and variables. It helps us to write clean, reusable and structured programming.

Class: A class is a user define data type that group variable (data) and function (method) together. Its like a blueprint to create objects.

Syntax:

```
class ClassName {
    access_specifiers :
    data members ;
    members_function ;
};
```

Object: An object is an instance of class.

```
void showBalance () {
        Cout << balance ;
    }
};
int main () {
    bank b1;
    b1.setBalance(5000);
    b1. showBalance () ;
    }.
```

- Output: 5000
- balance is private
- It cannot be access directly.
- we use function to access it.

b1.balance = 5000; (this will give error)

Problem statement: Bank Account using OOP.

① Public Acess specifier.

Member declared as public acess specifiers can be access anywhere in the program.

```
#include <iostream>
using namespace std;
class car {
    public:
        int speed;
        void show() {
            cout <<speed;
        }
    };
int main () {
    car c1;
    c1.speed =120;
    c1. show() ;
}.
```

- here speed is public so it can be accessed in main().

② Private Acess specifiers.

Members declared as private can only be acess inside the class.

they cannot be acess directly from outside the class.

Ex

```
#include <iostream>.
using namespace std;
class bank {
    private:
        int balance;
    public:
        void. setBalance (int b) {
            balance = b ;
        }
```